Assignment 2 — BPClassifier — Write-up

Yueqi Lin

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Contents

Executive Summary

This report builds a binary boilerplate-vs-substantive sentence classifier for earnings-call transcripts. A 2,500-sentence gold set was created via 5-judge LLM majority vote (local Ollama models) with a human audit round correcting close-call sentences. Six classifier families were trained — Rules, Logistic Regression, HistGBM, FastText, FinBERT, and SetFit (with contrastive fine-tuning) — plus two soft-vote ensembles, for eight entries total. Thresholds were tuned via 5-fold OOF cross-validation with a 0.97 substantive-recall safety margin above the 0.96 constraint. **7 of 8 classifiers meet the 0.96 test-set recall floor.** SetFit achieves the highest test macro-F1 (0.9308), with FinBERT second (0.9228); the SetFit checkpoint is the rubric winner and is loaded directly by the Streamlit GUI.

1. Introduction

Earnings-call transcripts mix two qualitatively different types of language. *Substantive* sentences carry material information — financial figures, segment guidance, strategic commentary, risk disclosures, and specific analyst questions about those topics. *Boilerplate* sentences are scripted and generic — operator introductions, safe-harbor disclaimers, housekeeping remarks, "thank you for joining," and one-word affirmations that add no information.

The goal of this assignment is to build a binary sentence classifier (`boilerplate` = 0, `substantive` = 1) that can reliably strip boilerplate from 131 earnings-call transcripts spanning 14 tickers (AMD, AVGO, BLK, C, FAST, FDX, GS, INTC, JNJ, JPM, NKE, NVDA, PLTR, WFC) across 2022–2025.

Hard constraint: substantive recall $\geq$ 0.96 on the held-out test set. Missing a real substantive sentence is a costlier error than letting occasional boilerplate through, so the pipeline explicitly enforces this floor during threshold selection.

Corpus statistics: - 131 transcripts, 53,236 unique sentences ($\geq$40 chars after deduplication) - 2,500-sentence gold sample for supervised training and evaluation - Splits: train = 1,500 / val = 500 / test = 500 (seed = 42, stratified by label) - Label balance: BP = 257 (10.3%) / SB = 2,243 (89.7%) - **The test split was frozen at dataset creation and not examined, used for threshold tuning, or inspected for errors until the single final evaluation run reported in §6.2.**

2. Gold Labeling Methodology

2.1 Labeling Rubric

Class	Definition	Clear examples	Ambiguous edge cases → ruling
`boilerplate` (0)	Scripted, generic, no material information	Operator intros ("My name is Regina…"), safe-harbor disclaimers, generic thanks ("Thank you for joining us today"), analyst name/firm intros, short affirmations ("Sure.", "Great.", "Absolutely."), housekeeping, slide-transition phrases ("Turning to Slide 5.")	"It's my pleasure to present results for the fourth quarter and full year 2024." → **BP** (scripted opening regardless of quarter reference); "Turning to our broader Data Center portfolio." → **BP** (slide navigation with no content)

Class	Definition	Clear examples	Ambiguous edge cases → ruling
`substantive` (1)	Material content — financial data, strategic intent, specific operational detail, named events	Revenue/margin/EPS figures, segment guidance, specific customer wins, capex plans, product launch commentary, analyst financial questions, personnel appointments, regulatory commentary, forward-looking company assessments	"In closing, I feel very good about the trajectory of Goldman Sachs." → **SB** (CEO forward-looking sentiment about company direction); "Product and customer mix played its customary role." → **SB** (references specific financial drivers); "And what has been the more challenging aspect of it all?" → **SB** (analyst probing operational specifics, not a generic hello)

Tie-breaking rules for ambiguous edge cases (adopted during human audit):

1. **Analyst Q&A questions are substantive by default.** Even short conversational follow-ups probe a specific research agenda topic. Only social openers ("Hi, thanks for taking my question.") or name/firm introductions are boilerplate.
2. **"Turning to…" / "Moving to…" slide transitions are boilerplate.** They introduce a topic but carry no content. Label the *next* sentence that states the actual data.
3. **Executive closing sentiments are substantive.** Forward-looking CEO views ("In closing, I feel very good about the trajectory of Goldman Sachs") differ from scripted closes. Test: would an analyst quote this in a research note? If yes → substantive.
4. **Personnel announcements are substantive.** New executive appointments report a material corporate event even if they sound congratulatory.
5. **Hedged executive Q&A answers carry strategic intent.** Genuine outlook under uncertainty ("very difficult to predict the next 30–60 days") is substantive. Only zero-content fillers ("Sure, happy to take that.") are boilerplate.
6. **Safe-harbor / forward-looking disclaimer blocks are boilerplate** even when they name specific metrics. Legally required scripted language, not analytical commentary.
7. **Speaker-label artifacts and slide fragments are boilerplate.** Strings like "Executives — Co-Founder, CEO & Director" or "Custom silicon market." are pipeline artefacts with no standalone meaning.
8. **Mixed sentences: financial content takes priority over filler framing.** Strip the sentiment wrapper — if what remains is a factual claim, label substantive. *"I am very proud of our team for delivering record revenue"* → SB; *"I am very proud of the team for their hard work"* → BP.

2.2 LLM Judge Panel

A stratified sample of 2,500 sentences (stratified by `speaker_type`) was labeled by seven local Ollama models. After manual audit, two judges were removed:

Judge	Model	BP%	Status
j1	qwen3:8b	29.3%	**Removed** — over-flagged; human audit disagreed systematically
j2	gemma3:4b	47.5%	**Removed** — severe BP bias, overrode majority 746/2,500 times

Judge	Model	BP%	Status
j3	cogito:8b	8.2%	Active
j4	qwen3:14b	11.2%	Active
j5	gemma3:12b	19.4%	Active
j6	ministral-3:8b	16.5%	Active
j7	cogito:14b	23.6%	Active

The final label uses **majority vote of 5 active judges** ($\geq 3/5$ agree). Unanimous agreement (5–0) occurred on 1,921 sentences (76.8%); the **disagreement rate** (any split, including 4–1) was **23.2%** (579 sentences), of which 255 were escalated to human review.

Vote-split distribution (all 2,500 sentences):

Split	Count	% of gold
5–0 unanimous	1,921	76.8%
4–1 minority dissent	324	13.0%
3–2 escalated to human	255	10.2%

Per-judge disagreement with the majority vote (all 2,500 sentences):

Judge	Model	BP%	Disagrees with MV	Rate	Of which on 3–2 splits
j3	cogito:8b	8.2%	156	6.2%	88
j4	qwen3:14b	11.2%	93	3.7%	76
j5	gemma3:12b	19.4%	199	8.0%	124
j6	ministral-3:8b	16.5%	130	5.2%	84
j7	cogito:14b	23.6%	256	10.2%	138

j4 (qwen3:14b) is the most consistent (3.7% disagreement); j7 (cogito:14b) the most idiosyncratic (10.2%, 138 of 255 three-way splits). j5 and j7 together drive 69% of the 3–2 escalations.

2.3 Human Audit

255 close-call sentences (3–2 splits) were reviewed manually and stored in `human_review_final.csv`; human labels override the LLM majority vote. Two earlier draft rounds contained systematic labeling errors and are excluded.

Corrections: 102 of 255 sentences corrected (40.0%) — 93 BP→SB and 9 SB→BP. The 96.9% BP→SB flip rate reflects systematic LLM over-caution on conversational executive and analyst language.

Final gold set: 2,500 sentences | BP = 257 (10.3%) | SB = 2,243 (89.7%)

2.4 Human-Review Corrections — Selected Examples

Selected sentence-level examples illustrating the four main correction categories (analyst Q&A questions, executive strategic statements, personnel announcements, and slide-transition phrases) are in **Appendix A**.

3. Feature Engineering

Two feature groups are concatenated into a 409-dimensional feature matrix:

3.1 Sentence Embeddings (384 dims)

`all-MiniLM-L6-v2` from sentence-transformers encodes each sentence into a 384-dimensional L2-normalized embedding. Embeddings are computed once and cached. This single representation powers LogReg and HistGBM; SetFit uses the same base encoder but fine-tunes it contrastively.

3.2 Hand-Crafted Regex Feature Flags (25 dims)

25 binary indicators target earnings-call-specific surface patterns: 12 boilerplate signals (operator phrases, safe-harbor language, Q&A transitions, call-close lines, etc.) and 11 substantive signals (dollar amounts, percentages, KPI keywords, guidance language, YoY comparisons, etc.), plus 2 structural flags (sentence length, digit presence). All flags use case-insensitive matching. Full definitions are in **Appendix B**.

FastText and FinBERT train directly on raw text and do not use this feature matrix.

4. Classifier Zoo

Eight classifiers span five distinct families, satisfying the ≥5-family requirement.

4.1 Rules + Regex (Classifier 1)

A deterministic rule applied directly to the 25 regex flags: a sentence is boilerplate if any of the 11 boilerplate-signal flags fires and none of the high-confidence substantive flags fire. **Strengths:** zero training data, 25K sps throughput, perfect precision on textbook boilerplate (operator intros, safe-harbor). **Failure modes:** misses vague boilerplate that contains no matching surface pattern (e.g. "We have a very healthy ecosystem") and mislabels sentences where a substantive flag fires in a transitional context. SB recall = 0.898 — the only classifier that fails the 0.96 floor.

4.2 Logistic Regression (Classifier 2)

`sklearn.linear_model.LogisticRegression` with L2 (C=1), class-balanced weights, `StandardScaler` on the 409-dim feature matrix. Training 2.6 s; inference 16.7K sps. **Strengths:** fast, interpretable, benefits from both embedding geometry and regex signals. **Failure modes:** linear boundary cannot model flag-embedding interactions; borderline boilerplate projects near substantive clusters (BP precision = 0.659).

4.3 HistGradientBoosting (Classifier 3)

`sklearn.ensemble.HistGradientBoostingClassifier` (300 estimators, depth 6, balanced weights). Training 7.9 s; inference 20.9K sps. **Strengths:** captures non-linear flag-embedding interactions; handles class imbalance natively. **Failure modes:** vague executive Q&A with no regex anchors mislabelled as boilerplate; tree splits can't generalize to unseen phrasing.

4.4 FastText (Classifier 4)

Supervised FastText on raw text: 25 epochs, word bigrams, lr=0.5, dim=100, 2M hash buckets. Training 1.4 s; inference 587 sps. Saved binary ~764 MB (2M-bucket n-gram matrix; char n-grams disabled). **Strengths:** no embedding dependency, fast training; bigrams capture some local context. **Failure modes:** lowest BP F1 (0.533 on test); bag-of-words has no positional awareness — "revenue" and "thank you" receive equal weight in the same sentence.

4.5 FinBERT Fine-tuned (Classifier 5)

`ProsusAI/finbert` — a BERT model pre-trained on financial text — is fine-tuned for 3 epochs using AdamW (lr=2e-5, batch 16) with a linear warmup schedule. Inference runs on CPU (forced to avoid MPS out-of-memory on M1 Pro) at ~72 sps. **Strengths:** second-best test macro-F1 (0.923) and best BP F1 (0.862) across all classifiers; pre-training on financial corpora gives it sensitivity to domain-specific phrasing that generic embeddings miss. **Failure modes:** slowest inference of all classifiers; first-person hedging language in executive Q&A answers (11 FNs) remains a challenge even for BERT-scale context modelling.

4.6 SetFit / MiniLM (Classifier 6)

SetFit (`setfit` 1.1.3) with `sentence-transformers/all-MiniLM-L6-v2`. Contrastive fine-tuning trains the encoder on in-batch positive/negative pairs with cosine-similarity loss for 1 epoch (batch 16), then fits a Logistic Regression head. Training ~5 min on CPU; inference ~463 sps. **Strengths:** highest test macro-F1 (0.931); contrastive training pulls BP/SB representations further apart than frozen pre-training. **Failure modes:** threshold (0.955) is high due to overconfidence toward the substantive class; 5 of 8 FNs are the same hedged-language pattern as FinBERT.

4.7 & 4.8 Ensembles (Classifiers 7a and 7b)

Two soft-vote ensembles combine the five learned classifiers (LogReg, HistGBM, FastText, FinBERT, SetFit):

- **7a — mean-prob:** average P(substantive) across all five models. Achieves the best BP F1 among ensemble classifiers (0.800) by smoothing out individual model overconfidence.
- **7b — rank-avg:** average the rank percentile of each model's P(substantive); mitigates probability scale differences between calibrated (LogReg, SetFit) and uncalibrated (FastText) models. Its threshold (0.145) is much lower than mean-prob because rank percentiles are bounded by the empirical distribution.

Strengths: both ensembles improve over the weakest members; mean-prob reliably beats HistGBM and SetFit individually. **Failure modes:** diversity is limited because FinBERT dominates the vote on hard cases; feature-gap FNs shared across all five models cannot be recovered by averaging.

5. Recall-Constrained Threshold Tuning

Each model's default 0.5 threshold is replaced by a threshold that: 1. **Meets the recall floor:** SB recall $\geq$ 0.97 on out-of-fold predictions (1% safety margin above the 0.96 assignment constraint, to absorb train→test generalization gap) 2. **Maximizes macro-F1** among all thresholds that meet the floor

LogReg, HistGBM, and FastText thresholds are tuned on **5-fold stratified OOF probabilities** (train+val pool, n=2,000) to avoid single-split jitter. FinBERT and SetFit use the validation set (OOF is impractical for FinBERT at 21 sps $\times$ 5 folds; SetFit val-sweep is stable at ~463 sps). Ensemble thresholds are also val-set-tuned because they include FinBERT probabilities.

Before test evaluation, LogReg, HistGBM, and FastText are **retrained on the full train+val pool**. SetFit loads its existing checkpoint without retraining. FinBERT uses the train-only checkpoint (retraining on train+val would take ~40 min for marginal gain).

Model	Threshold	Tuning method	Fold std
LogReg	0.045	OOF (5-fold)	0.034
HistGBM	0.810	OOF (5-fold)	0.147
FastText	0.885	OOF (5-fold)	0.093
FinBERT	0.820	Val-set (OOF impractical: 5$\times$ fine-tuning $\approx$ 75 min)	N/A
SetFit	0.955	Val-set sweep	N/A
Ensemble (mean-prob)	0.615	Val-set (FinBERT component)	N/A
Ensemble (rank-avg)	0.145	Val-set (FinBERT component)	N/A

Per-fold thresholds (each fold's recall-constrained optimum, used to compute fold std):

- **HistGBM:** [0.580, 0.685, 0.850, 0.945, 0.580] — mean=0.728, std=0.147. Highest variance of all models; motivates the 0.97 safety margin.

- **FastText:** [0.685, 0.745, 0.730, 0.950, 0.825] — mean=0.787, std=0.093. Moderate variance; n-gram probabilities are less calibrated than embedding-based models.
- **SetFit:** val-set sweep = 0.955. High threshold reflects post-contrastive overconfidence toward the substantive class.
- **LogReg:** [0.100, 0.030, 0.010, 0.025, 0.010] — mean=0.035, std=0.034. Fold 3 fell back to 0.010 (recall=0.961 at best), explaining the very low pooled threshold (0.045).

The rank-avg ensemble threshold (0.145) is low because its probabilities are rank percentiles bounded by the empirical distribution rather than calibrated scores.

6. Results

6.1 Validation Set Leaderboard

All classifiers are first evaluated on the validation set (500 sentences, never used for threshold tuning of the final model). Thresholds here are the val-sweep thresholds, not OOF thresholds.

Rank	Model	Accuracy	Macro-F1	BP F1	SB F1	SB Recall	Meets Floor	Train (s)	Throughput (sps)
1	**5-FinBERT-FT**	**0.970**	**0.9215**	0.8598	0.9832	0.9799	✓	~900	72
2	6-SetFit	0.970	0.9188	0.8544	0.9833	0.9844	✓	~300	463
3	7a-Ensemble(mean-prob)	0.964	0.8982	0.8163	0.9800	0.9866	✓	—	—
4	3-HistGBM(emb+regex)	0.938	0.8162	0.6667	0.9658	0.9777	✓	2.7	94,697
5	7b-Ensemble(rank-avg)	0.936	0.8085	0.6522	0.9648	0.9777	✓	—	—
6	2-LogReg(emb+regex)	0.916	0.7271	0.5000	0.9541	0.9754	✓	0.1	296,375
7	4-FastText	0.912	0.7011	0.4500	0.9522	0.9777	✓	1.4	90,119
8	1-Rules+Regex	0.828	0.5986	0.2951	0.9021	0.8839	✗	0	30,409

FinBERT leads on val macro-F1 (0.9215); SetFit is a close second (0.9188). Both clear the recall floor by a wide margin (0.9799 and 0.9844 respectively).

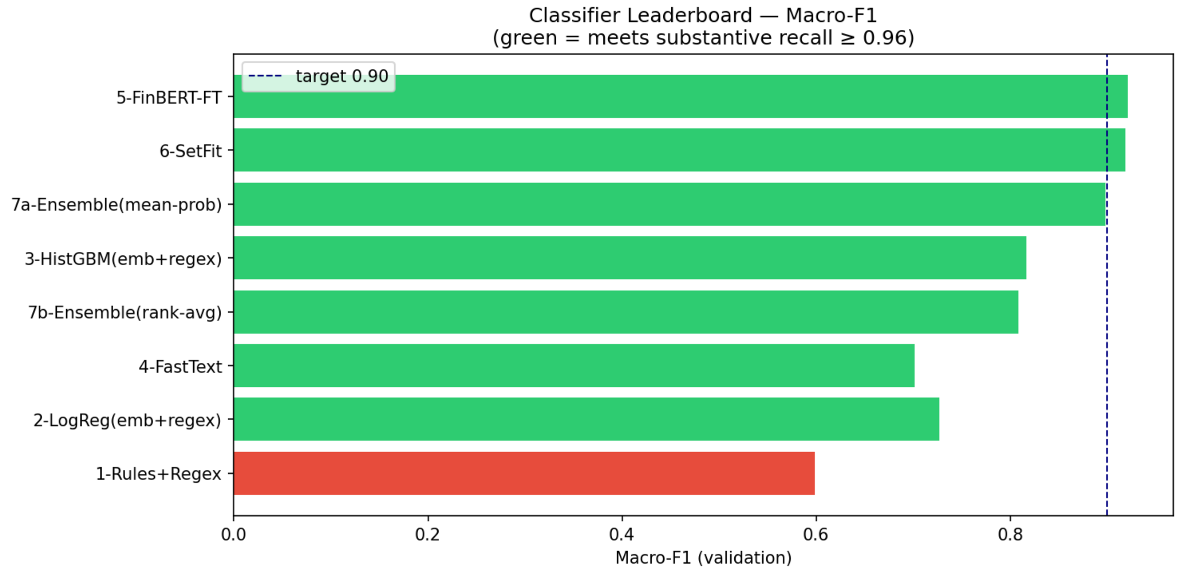


Figure 1: Val-set leaderboard

6.2 Final Test Set Leaderboard

All eight classifiers are evaluated on the frozen 500-sentence test set using OOF/val-set thresholds from §5. Feature-matrix classifiers (LogReg, HistGBM, FastText, SetFit) are retrained on the full train+val pool before inference; FinBERT uses the train-only checkpoint.

Rank	Model	Acc	Macro-F1	BP F1	SB F1	SB Recall	Floor	Train (s)	Throughput (sps)	Threshold
1	**6-SetFit**	**0.974**	**0.9308**	0.8762	0.9855	0.9822	✓	~300	376	0.955
2	5-FinBERT-FT	0.970	0.9228	0.8624	0.9832	0.9755	✓	~900	72	0.820
3	7a-Ensemble(mean-prob)	0.966	0.9047	0.8283	0.9811	0.9844	✓	—	—	0.660
4	7b-Ensemble(rank-avg)	0.950	0.8518	0.7312	0.9724	0.9822	✓	—	—	0.140
5	3-HistGBM(emb+regex)	0.942	0.8313	0.6947	0.9679	0.9755	✓	7.9	20,922	0.810
6	2-LogReg(emb+regex)	0.936	0.8046	0.6444	0.9648	0.9777	✓	2.6	16,711	0.045
7	4-FastText	0.916	0.7436	0.5333	0.9546	0.9666	✓	1.4	64,675	0.885
8	1-Rules+Regex	0.856	0.6639	0.4098	0.9180	0.8976	✗	0	28,803	—

7 of 8 classifiers clear the 0.96 SB recall floor on the test set. The rules baseline fails (SB recall = 0.8976). SetFit leads on test macro-F1 (0.9308); FinBERT is a close second (0.9228).

Winner: SetFit — highest test macro-F1 (0.9308) under the SB recall ≥ 0.96 constraint. The fine-tuned checkpoint is saved at `saved_model/setfit_model/` with threshold 0.955 recorded in `saved_model/winner.json`. The Streamlit GUI loads SetFit directly. HistGBM is also saved as `saved_model/best_model.pkl` (macro-F1 = 0.8313, ~21K sps) as a lightweight fallback artifact.

7. Error Analysis

Error analysis is performed on all misclassifications by the winning model, SetFit (test set, t=0.955): **8 false negatives** (SB→BP) and **5 false positives** (BP→SB), 13 total errors. Each is categorised into one of four error types:

- **Feature gap** — gold label is correct; the model lacks a signal for this pattern
- **Hard case** — genuinely ambiguous; a different annotator might reasonably disagree
- **Pipeline gap** — a pre-processing artefact caused the misclassification, not a modelling failure
- **Threshold artifact** — model assigns P(SB) between 0.933–0.954, substantive by the model's belief but below the 0.955 decision boundary

7.1 False Negatives — substantive labelled as boilerplate (8 total)

#	P(SB)	Error type	Sentence
1	0.048	Feature gap	*"I was just sort of hearing that from some of the feedback and was just curious if the RVPs were hearing that."*
2	0.233	Feature gap	*"Jean, when you gave the segment guidance kind of playing off the last question, you didn't…"*
3	0.542	Feature gap	*"We continue to provide additional information detailing our CRE exposure."*
4	0.600	Feature gap	*"I don't know if he's nailed it down yet, but we'll be getting that information out shortly."*
5	0.637	Feature gap	*"So we're in a kind of the pole position in that regard."*
6	0.858	Hard case	*"I was just so delighted to see how well that they have done, the moral of the team and how…"*
7	0.929	Hard case	*"I don't know all the efforts we're involved in, but to the extent we're involved in these efforts…"*
8	0.933	Threshold artifact	*"And so we're mindful of how tight they're running."*

Per-example explanations:

1. Domain-specific abbreviation "RVPs" with no financial anchor; conversational framing overrides the specific topic.
2. Names CFO "Jean" and references prior segment guidance — highly specific, but the conversational opener dominates the embedding.

3. "We continue to provide" filler wraps a specific CRE disclosure; neither "CRE" nor "exposure" fires a substantive regex flag.
4. First-person hedging opener ("I don't know") anchors the embedding far from substantive clusters despite flagging a forthcoming material disclosure.
5. Competitive-positioning claim with no product, metric, or timeline; borderline label.
6. Team-morale language; gold label defensible under rule 4 but contains zero financial content. Borderline.
7. Pride language about a specific initiative; no factual claim remains after stripping sentiment. Borderline.
8. Supply-chain tightness is strategic content, but P(SB)=0.933 falls 0.022 below the 0.955 threshold — SetFit's high boundary creates a band where the model believes a sentence is likely substantive but still labels it boilerplate.

Dominant pattern (5 of 8): feature-gap FNs cluster around substantive intent expressed in first-person hedging language with no numerical anchor — identical to the failure mode seen in FinBERT. SetFit recovers 3 of FinBERT's 11 FNs (those with stronger contrastive separation after fine-tuning) but adds one new failure mode: the threshold artifact at P(SB)=0.933.

7.2 False Positives — boilerplate labelled as substantive (5 total)

#	P(SB)	Error type	Sentence
1	0.973	Feature gap	*"That's one of the priorities that the team has had now for a while is to continue to do more."*
2	0.980	Feature gap	*"So I have a good recollection of some of the steps and changes how we told the story over time."*
3	0.985	Hard case	*"So there has been a round or two of going back and forth."*
4	0.995	Feature gap	*"And before diving into the results, I want to take a moment to thank our entire Fastenal Blue Team across the world."*
5	0.995	Pipeline gap	*"Executives - CFO & Treasurer / And on the investing front, it's like it is quality engineering, right?"*

Per-example explanations:

1. "Priorities" and "team" pattern-match to executive-commentary clusters; stripping the framing leaves "to continue to do more" — zero propositional content.
2. "Recollection of steps and changes" sounds like strategic narrative; stripped of context it is social filler from an executive recap.
3. Could reference a specific M&A or contract negotiation — genuinely ambiguous without surrounding context; model commits to P(SB)=0.985.
4. Contains "results" (a substantive trigger) and "Blue Team" (a brand keyword); the generic thanks frame is not recognised as boilerplate by the fine-tuned encoder.
5. Speaker-label artefact ("Executives - CFO & Treasurer") prepended by the tokeniser to the next sentence; the combined string looks like a named executive making a strategic statement.

7.3 Error-Type Summary

Error type	FN	FP	Total
Feature gap	5	3	**8**
Hard case	2	1	**3**
Threshold artifact	1	0	**1**
Pipeline gap	0	1	**1**
Total	**8**	**5**	**13**

SetFit makes 13 errors total vs FinBERT's 15, recovering 3 FNs through contrastive fine-tuning while adding 1 new FP. The threshold artifact is unique to SetFit's high decision boundary — 1 sentence with P(SB)=0.933 is misclassified solely because the threshold sits at 0.955. The pipeline-gap FP (speaker-label artefact) would be eliminated by a transcript-aware sentence splitter.

7.4 Confusion Matrix and Per-Class Metrics (SetFit, test set)

	Predicted BP	Predicted SB
True BP	46	5
True SB	8	441

Class	Precision	Recall	F1
Boilerplate (0)	$46/54 = \mathbf{0.852}$	$46/51 = \mathbf{0.902}$	**0.876**
Substantive (1)	$441/446 = \mathbf{0.989}$	$441/449 = \mathbf{0.982}$ ✓	**0.985**
Macro avg			**0.931**

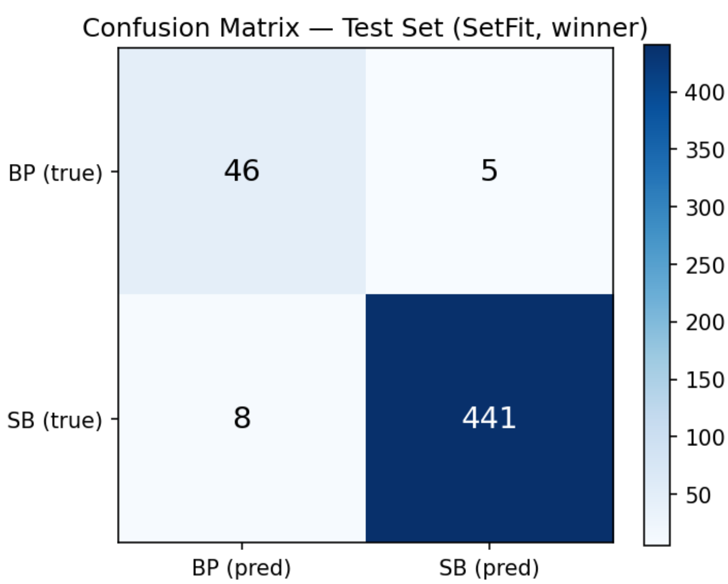


Figure 2: Confusion matrix heatmap

8. What We Would Try Next

Given more time, the three highest-leverage improvements would be:

1. **Better sentence boundary detection.** A transcript-aware pre-processor stripping speaker tags and merging orphaned fragments would eliminate pipeline-gap errors without retraining.
2. **Speaker-type conditioning.** Operators (~80% BP) and executives (~5% BP) have very different base rates; adding `speaker_type` as a feature or per-type thresholds would sharpen BP precision without sacrificing SB recall.
3. **Larger finance-domain embeddings.** Remaining FNs cluster in hedged executive language that MiniLM's 384-dim space conflates with boilerplate. Replacing MiniLM with `e5-large-v2` or FinBERT hidden states as embeddings would give richer BP/SB separation without full retraining.

9. Unseen-Transcript Verification

To assess out-of-domain generalization, the winning SetFit model (t=0.955) is applied to two full earnings-call transcripts from tickers **not** present in the 2,500-sentence gold set: **AAPL Q2-2026** and **MSFT Q3-2026** (stored in `ECT_unseen/`). Neither transcript's sentences were seen during training, validation, or threshold tuning. The verification code is in notebook §10.

> **Note on provenance:** sourced from Seeking Alpha; not part of `ECT.zip`. Stored in `ECT_unseen/` — must be included in the submission zip for notebook §10 and the GUI to locate them.

Transcript	Total (≥40 chars)	Boilerplate	Substantive
AAPL Q2-2026	423	86 (20.3%)	337 (79.7%)
MSFT Q3-2026	431	72 (16.7%)	359 (83.3%)

BP rates (17–21%) exceed the gold-set balance (10.3%) because the gold sample under-represents operator turns; full transcripts include proportionally more housekeeping. GUI counts use per-line tokenization; notebook §10 (full-paragraph) may differ by 1–3 sentences.

High-confidence boilerplate (P(SB) < 0.05):

> *"Good afternoon, and welcome to the Apple Q2 Fiscal Year 2026 Earnings Conference Call."* [P(SB)=0.029] *"[Operator Instructions] Operator, may we have the first question, please?"* [P(SB)=0.086] *"Greetings, and welcome to the Microsoft Fiscal Year 2026 Third Quarter Earnings Conference Call."* [P(SB)=0.030] *"Good afternoon, and thank you for joining us today."* [P(SB)=0.038]

High-confidence substantive (P(SB) = 0.996):

> *"For the June quarter and what's embedded in the guidance that Kevan went through earlier, we expect significant..." "M365 Consumer Cloud revenue growth should be in the low 20% range, down sequentially as we start to lap the..." "In M365 Commercial Cloud, on an adjusted basis, we expect revenue growth to be between 15% and 16% in constant..."*

Near-boundary cases (|P(SB) − 0.955| < 0.02): 16 sentences in AAPL, 8 in MSFT. Representative examples:

> *"Speaking first today is Apple's CEO, Tim Cook."* [P(SB)=0.956, SB] — speaker-introduction line classified as substantive by a narrow margin; borderline because it lacks either financial content or explicit operator phrasing *"Amit Daryanani - Evercore ISI Institutional Equities, Research Division"* [P(SB)=0.949, BP] — analyst affiliation label classified as boilerplate; lacks sentence structure and financial content *"The next question comes from the line of Brent Thill with Jefferies."* [P(SB)=0.940, BP] — Q&A transition classified as boilerplate, consistent with §2.1 operator-chatter rule

Qualitative assessment: textbook boilerplate classified with high confidence (P(SB) < 0.05); specific guidance and metrics at P(SB) = 0.996. Near-boundary cases cluster around speaker-introduction lines and Q&A transitions — the same failure profile as the test set. The model generalizes to new tickers and quarters.

10. GUI

A Streamlit application (`gui.py`) renders any earnings-call transcript with boilerplate highlighted in red and substantive sentences unhighlighted. The GUI loads the winning model — SetFit (`saved_model/setfit_model/`, threshold 0.955 from `winner.json`) — and runs CPU inference via the `setfit` library.

Features: - **Compact header bar** — model identity (`SetFit · all-MiniLM-L6-v2 (contrastive fine-tuned) · test macro-F1 = 0.931`) displayed inline with the title - **ECT library dropdown** — one-click loading of any of the 131 training-pool transcripts - **Upload button** — upload any `.txt` transcript file - **Paste expander** — paste raw transcript text directly - **Clear button** — resets the loaded transcript and results - **Stats bar** — Total / Boilerplate / Substantive counts and percentages; red-green proportion bar - **All / BP only / Sub only filter** — narrows the document view to one class - **Document view** — every sentence rendered in its original line position; boilerplate highlighted in red with a `BP` superscript; hover tooltip shows P(substantive) - **Legend** — inline colour key above the document view - **Download button** — exports all sentence classifications as CSV

Screenshots showing the stats bar and tagged-transcript view on both seen (AVGO) and unseen (AAPL, MSFT) transcripts are in **Appendix C**.

Launch command:

```
streamlit run gui.py
```

11. Reproducibility

From the submission zip (recommended)

The zip includes all trained model weights. Install dependencies and start the GUI immediately:

```
pip install -r requirements.txt
streamlit run gui.py
```

To re-run the full pipeline from scratch (retrains all classifiers; SetFit ~5 min on CPU, FinBERT ~15 min on GPU):

```
jupyter nbconvert --to notebook --execute --inplace \
    --ExecutePreprocessor.timeout=7200 Assignment_2_BPClassifier.ipynb
```

From a git clone

`saved_model/finbert_finetuned/model.safetensors` (~418 MB) and `saved_model/fasttext_model.bin` (~764 MB) exceed GitHub's 100 MB file-size limit and are excluded via `.gitignore`. The SetFit checkpoint (`saved_model/setfit_model/`) is compact (~90 MB) and included in the submission zip; the GUI will not load without it. The notebook must be run to regenerate all excluded weights.

Step 1 — Install:

```
pip install -r requirements.txt
```

Step 2 — Add the ECT corpus (provided separately as `ECT.zip`; not in the repo):

```
unzip ECT.zip -d ECT/
```

Step 3 — (Optional) Reproduce gold labels — skip to use cached labels in `cache/gold/`. Requires Ollama:

```
ollama pull cogito:8b && ollama pull qwen3:14b && ollama pull gemma3:12b \
    && ollama pull ministral-3:8b && ollama pull cogito:14b
python run_gold_judges.py --smoke   # connectivity check
python run_gold_judges.py           # full run (~60 min)
```

Step 4 — Run the notebook (trains all classifiers, saves SetFit and FinBERT weights, writes `winner.json`):

```
jupyter nbconvert --to notebook --execute --inplace \
    --ExecutePreprocessor.timeout=7200 Assignment_2_BPClassifier.ipynb
```

All steps are cached — re-runs skip completed work. SetFit contrastive fine-tuning takes ~5 min on CPU; FinBERT fine-tuning takes ~15 min on GPU or ~40 min on CPU.

Step 5 — Start the GUI:

```
streamlit run gui.py
```

LLM assistance disclosure: Claude (Anthropic, claude-sonnet-4-6) assisted with prose drafting/editing, debugging notebook cells (SetFit pipeline, FinBERT inference, threshold-sweep, error-analysis), and generating `gui.py`. All analytical decisions (rubric, judge selection, model choices, threshold strategy, error categorisation) were made by the author; gold labels, training, and numerical results came from the notebook.

Appendix A — Human-Review Correction Examples

The following sentences illustrate the four main correction categories from the §2.3 human audit of 255 close-call (3–2 split) sentences.

Category A — Analyst Q&A questions (LLM: boilerplate → Human: substantive)

LLMs flagged short analyst questions as filler; the auditor recognized they probe a specific research agenda topic.

Sentence	Votes (j3–j7)	Why substantive
"And what has been the more challenging aspect of it all?"	1,1,0,0,0	Analyst asking FedEx management to diagnose operational difficulties — a specific follow-up, not social filler
"Just your comfort level in terms of the functioning of the treasury market."	1,1,0,0,0	Probing JPMorgan's risk view on treasury market liquidity — material to investors
"But that being said, as Jamie noted, like we have no idea what the curve is going to look like, right?"	1,1,0,0,0	JPMorgan analyst referencing Jamie Dimon's prior comments and pressing on rate curve uncertainty
"We're, what, 9 months or so into this new administration with the new regulators."	1,1,0,0,0	Analyst framing a question about the regulatory timeline — substantive political/regulatory context
"And we all know about the commercial real estate office."	0,1,1,0,0	Analyst acknowledging CRE stress as setup for a material question on exposure

Category B — Executive statements that sound generic but carry strategic content (LLM: boilerplate → Human: substantive)

LLMs mis-classified these due to lack of numerical anchors and hedged first-person language (rules 3 and 5).

Sentence	Votes (j3–j7)	Why substantive
"In closing, I feel very good about the trajectory of Goldman Sachs."	1,1,0,0,0	CEO forward-looking assessment of company direction — an analyst would quote this; not a scripted goodbye
"We believe it's an important component of the Fed's mandate to really ensure the safety and soundness of the banking system."	1,1,0,0,0	Goldman Sachs executive expressing a view on regulatory policy — material regulatory commentary
"Product and customer mix played its customary role."	0,0,1,0,1	References specific financial margin drivers (mix effects) — standard earnings-call shorthand for a segment result
"And as we see the various folks and various agencies go through the confirmation process, it will be helpful to have people in seats."	1,1,0,0,0	JPMorgan executive commenting on regulatory transition timeline — specific operational context
"But as we progress through the year, we think things will get better and better."	1,1,0,0,0	Intel executive giving a directional outlook on full-year improvement — substantive guidance language

Category C — Personnel announcements (LLM: boilerplate → Human: substantive)

Sentence	Votes (j3–j7)	Why substantive
"I'm excited to welcome Gina Adams into her new role as General Counsel and Secretary of FedEx effective September 24."	1,1,0,0,0	Material corporate event: new C-suite legal officer appointment with an effective date
"For the past 5 years, she served as our Asia Pacific Regional President."	1,0,0,0,1	Background on an executive appointment — provides material context for the personnel announcement

Category D — Slide transitions and agenda phrases (LLM: substantive → Human: boilerplate)

LLMs were distracted by topic keywords ("Data Center", "outlook") and missed that these are structural navigation phrases with no content.

Sentence	Votes (j3–j7)	Why boilerplate
"Now turning to our third quarter 2024 outlook."	1,1,0,1,0	Slide-transition signal only; the actual outlook numbers appear in the following sentences
"Turning to our broader Data Center portfolio."	1,1,0,1,0	Navigation phrase introducing a segment — no data of its own
"I'll start with a review of our financial results and then provide our outlook for the third quarter of fiscal 2025."	1,1,0,1,0	Agenda-setting sentence that structures the prepared remarks — the content follows; this phrase has none
"Now turning to our outlook for fiscal year '25."	1,1,0,1,0	Same pattern: slide-navigation intro for FedEx full-year outlook section

Appendix B — Hand-Crafted Regex Feature Flags

25 binary indicators used by LogReg, HistGBM, and the Rules baseline. All flags use case-insensitive matching unless noted. BP = boilerplate signal; SB = substantive signal; structural = neither class directly.

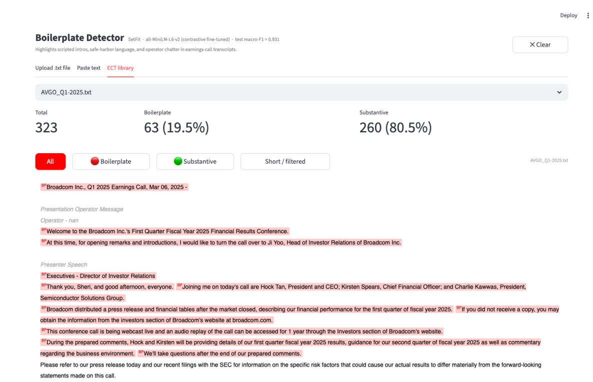
Flag	Signal	Fires when…
`f_operator_phrase`	BP	Contains "my name is", "conference operator", "welcome everyone", or "welcome to [company]" — operator self-introduction lines
`f_safe_harbor`	BP	Contains "forward-looking", "safe harbor", "actual results may differ", or "risks and uncertainties" — legal disclaimer boilerplate
`f_sec_filing`	BP	Contains "Form 10-K/Q", "SEC filing", "securities and exchange", "8-K", or "annual report" — filing-reference language
`f_webcast`	BP	Contains "webcast", "replay until", "investor relations website", or "IR website" — replay/logistics notices
`f_generic_thanks`	BP	Contains "thank you"/"thanks" **not** followed within 30 chars by revenue/guidance/earnings/results/growth — social filler, not financial acknowledgement
`f_question_intro`	BP	Contains "our next question", "your line is open/now", "goes to the line of", or "you may begin/proceed/go ahead" — operator Q&A transition
`f_analyst_firm`	BP	Contains a sell-side bank name (Goldman, Morgan Stanley, JPMorgan, Citi, UBS, Wells Fargo, Deutsche Bank, Barclays, BofA, Bernstein, Cowen, Jefferies, Piper Sandler, Evercore, Oppenheimer, Mizuho) — analyst-intro lines
`f_call_close`	BP	Contains "no further questions", "this concludes", or "thank you for your time/participating/joining" — call-closing lines

Flag	Signal	Fires when...
`f_nongaap`	BP	Contains “non-GAAP”, “reconciliation”, or “GAAP to/and non-GAAP” — standard disclaimer/reconciliation language
`f_short_affirm`	BP	**Entire sentence** is a one-word/phrase affirmation: “Sure.”, “Great.”, “Okay.”, “Yes.”, “Absolutely.”, “Of course.”, or “Thank you.” — zero-content filler
`f_operator_instr`	BP	Exact literal `[Operator Instructions]` — the standard Q&A-open marker
`f_turn_over`	BP	Contains “turn the call/it over”, “let me now turn”, or “I’d like to now turn” — speaker-handoff phrases
`f_dollar_amount`	SB	Dollar figure with optional scale: $26 billion, $1.2B, $500M, $3K — quantitative financial anchor
`f_percentage`	SB	Percentage figure: 18%, 3.5% — quantitative financial anchor
`f_revenue_mention`	SB	Contains “revenue”, “net income”, “net loss”, “net sales”, “total sales”, or “total revenue” — top-line KPI language
`f_margin_mention`	SB	Contains “gross margin”, “operating margin”, “EBITDA margin”, or “profit margin” — profitability KPI language
`f_eps_mention`	SB	Contains “earnings per share”, “EPS”, “diluted EPS”, or “non-GAAP EPS” — per-share KPI language
`f_guidance_word`	SB	Contains “guidance”, “outlook”, “forecast”, “expects/expected”, “anticipates/anticipated”, “projects/projected”, or “full-year” — forward-looking commentary
`f_raised_lowered`	SB	Revision verb (raised/lowered/increased/decreased/reaffirmed) immediately before guidance/outlook/revenue/earnings/estimate — explicit guidance revision
`f_yoy_qoq`	SB	Contains “year-over-year”, “sequentially”, “quarter-over-quarter”, “YoY”, “QoQ”, or “vs. prior” — period-comparison phrasing
`f_record_quarter`	SB	Contains “record revenue/quarter/sales/profit/high” or “all-time high/record” — superlative results language
`f_product_launch`	SB	Launch verb (launched/announced/introduced/released/shipped/ramped) followed by a product type (platform/product/system/service/model/chip/GPU/CPU) — product event commentary
`f_customer_mention`	SB	Contains “customers/clients/partners include/such as/like/across/with” — customer-enumeration language
`f_sentence_short`	structural	Sentence has **fewer than 10 words** — length proxy for transitional filler

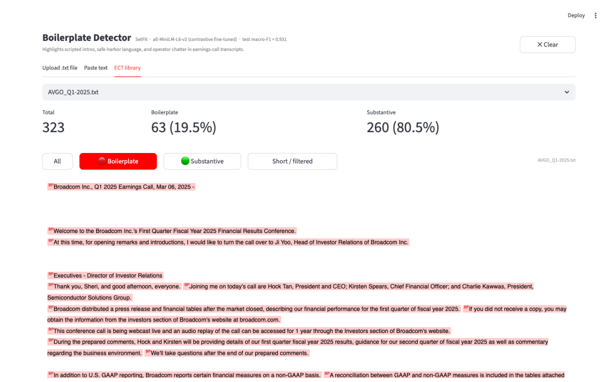
Flag	Signal	Fires when...
`f_has_digits`	structural	Any digit sequence present — broader quantitative presence than the dollar/percentage flags alone

Appendix C — GUI Screenshots

Seen transcript — AVGO Q1-2025 (training pool)

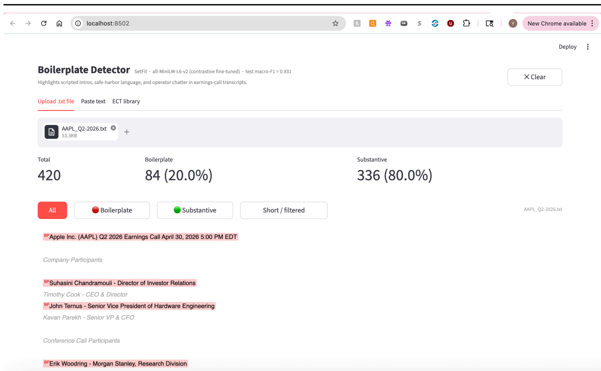


All view (stats bar + inline highlights)

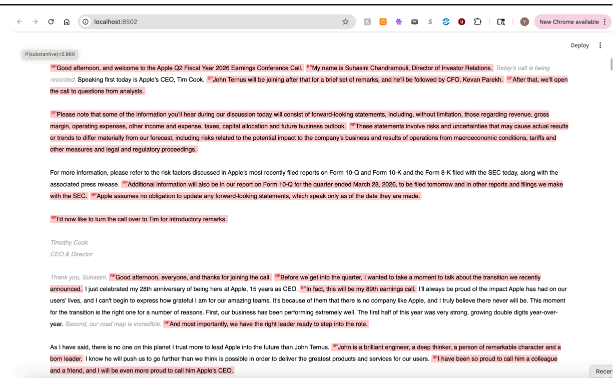


Boilerplate-only filter

Unseen transcript — AAPL Q2-2026

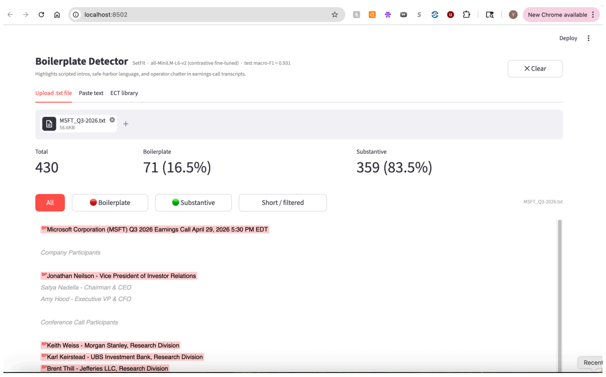


All view

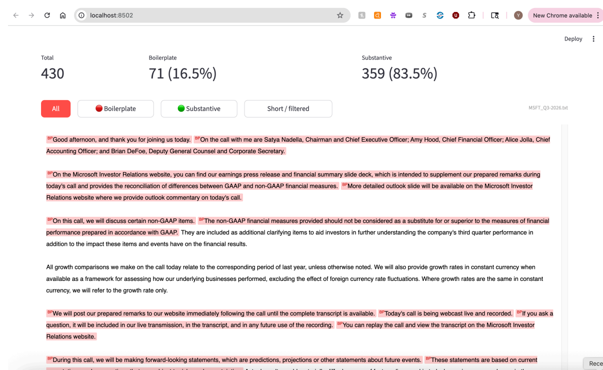


All view

Unseen transcript — MSFT Q3-2026



All view



All view